

Reading permutation runs from a plane tree

The original statistic and arbitrary run weights

EULER

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The original problem. For a permutation, let mnd be the sum of $\lfloor r/2 \rfloor$ over its maximal descending runs of lengths r , and define mna in the same way using ascending runs. For a rooted plane tree, let mark count its non-root vertices with a leaf child. Lin, Liu and Yan ask for a natural tree statistic st such that, for every $n \geq 1$,

$$\sum_{\pi \in S_n(231)} x^{\text{mnd}(\pi)} y^{\text{mna}(\pi^{-1})} = \sum_{T \in \mathcal{T}_n} x^{\text{mark}(T)} y^{\text{st}(T)}. \quad (1)$$

Here $\mathcal{T}_n$ consists of plane trees with n edges, and $S_n(231)$ consists of the permutations avoiding the pattern 231 [1, Problem 2.18]. In particular, the inverse on the left matters.

We will read an entire list of run lengths directly from a tree. Taking half of each length, rounded down, solves the original problem. Keeping the list instead of discarding it then gives the natural extension: every multiplicative weighting of the run lengths at once.

1 A reversible cut of a tree

Write $\bullet$ for a single vertex, and $P(F)$ for a root with ordered child forest F . Every other tree will be cut into a left tree L and a right tree R , with one fewer edge in total.

If the root has a leaf child, cut at the first such child. The children before it become the children of L , and the children after it become the children of R . Thus

$$T = P(F, \bullet, G) \quad \longmapsto \quad L = P(F), \quad R = P(G),$$

where F contains no leaf tree.

If the root has no leaf child, follow leftmost children to the first leaf. Let $F_0, \dots, F_m$ be the side forests encountered along this path, starting at the root and ending at the leaf's parent; $m \geq 1$. The side forest at each vertex consists of the children after the path child. In this case put

$$L = P(F_0, \bullet, F_m), \quad R = P(P(F_1), \dots, P(F_{m-1})).$$

In words, keep the two end forests together with the distinguished leaf, and hang the intermediate forests, in order, from a new root.

This cut is invertible. If L has no leaf child at its root, insert a leaf between the two child forests; that reverses the first case. Otherwise split L at its first leaf child. Its two forests are the end forests of the leftmost path. Each child of R supplies, in order, one intermediate side forest; reassemble that path and append the corresponding forest at each vertex. This reverses the second case. The first case is identified by the absence of a root leaf in L , and the second by its presence, so every ordered pair (L, R) is obtained once. In both cases

$$|T| = 1 + |L| + |R|.$$

Here and below size means number of edges.

2 The statistic and a complete solution of the original problem

Associate an ordered list $\lambda(T)$ of positive integers to the tree. The one-vertex tree has the empty list. For a nonempty tree, cut it into (L, R) , increase the first entry of $\lambda(R)$ by one, and append the resulting list to $\lambda(L)$. When $\lambda(R)$ is empty, replace it by the one-entry list (1) before appending. This is a terminating rule on the ordered tree alone, and the entries of $\lambda(T)$ sum to $|T|$. Define

$$\text{st}(T) = \sum_{r \text{ in the list } \lambda(T)} \left\lfloor \frac{r}{2} \right\rfloor.$$

For instance, the two-edge chain has list (1, 1), while the root with two leaf children has list (2). Their pairs (mark, st) are respectively (1, 0) and (0, 1).

Theorem 1. *There is an explicit bijection from $\mathcal{T}_n$ to $S_n(231)$ which sends $\text{mark}(T)$ to $\text{mnd}(\pi)$ and sends $\lambda(T)$ to the ordered list of ascending-run lengths of π^{-1} . Consequently (1) holds.*

Proof. Map the one-vertex tree to the empty permutation. If the two smaller trees map to α and β , and $k = |L| + 1$, set

$$\pi = (k, \alpha_1, \dots, \alpha_{k-1}, k + \beta_1, \dots, k + \beta_{|R|}). \quad (2)$$

All values in the first block are smaller than those in the second; inside the first block, the new largest value comes first. Hence the construction preserves avoidance of 231.

Conversely, if a 231-avoiding permutation begins with k , all values below k must precede all values above k . Otherwise k , a later larger value, and a still later smaller value form a forbidden pattern. Thus (2) is its unique decomposition. Recursively recovering the two trees and applying the inverse tree cut proves bijectivity.

We check the descending statistic first. In (2), only the initial descending run of α changes: its length increases by one. The boundary before the β block is ascending. The initial descending-run length is therefore one plus that of α . By induction its parity is odd exactly when the root of the corresponding tree has a leaf child: the two cases of the tree cut reverse this property between T and L .

Increasing a descending run of length r changes its contribution $\lfloor r/2 \rfloor$ by one exactly when r is odd. This agrees with what happens to marked vertices in the cut. In the first case, all non-root marked vertices retain their status. In the second case, the parent of the distinguished leaf was marked and non-root, but is absorbed into the root of L ; all other marked vertices are retained. Thus both statistics are the sum of the two smaller statistics, plus one exactly when the root of L has a leaf child. They agree by induction.

Finally, read positions in (2) in increasing value order:

$$\pi^{-1} = (\alpha^{-1} + 1), 1, (\beta^{-1} + k). \quad (3)$$

If the first block is present, its last entry exceeds 1, so a descent separates it from the middle entry. The middle 1 increases the first ascending run of the last block by one, or forms a singleton when that block is empty. This is exactly the rule defining $\lambda(T)$. Induction proves the assertion about the whole list, including both empty branches. Taking the sums of half-lengths and then summing over the bijection proves the original identity. $\square$

The tree cut and the original two statistics come from the released construction [2]. The preceding proof includes them in full; no permutation statistic was used to define the tree list.

3 Do not discard the run lengths

Give a run of length r a weight w_r , with the empty object having weight one. The w_r may be indeterminates, numbers, or zero. Record these choices in the single run series

$$W(s) = 1 + w_1s + w_2s^2 + \cdots, \quad W_+(s) = w_1 + w_2s + w_3s^2 + \cdots.$$

Let $F(z; x, u)$ count our trees by edges, marked vertices, and the first entry of their list, with each list entry also receiving its weight w_r . The empty tree has first entry zero. By Theorem 1, the same series counts 231-avoiding permutations by mnd, their first inverse ascending-run length, and all inverse ascending-run weights.

Theorem 2. *There is a unique series a with constant term zero satisfying*

$$a = z + xz^2W(a)W_+(a). \quad (4)$$

The complete weighted answer is

$$\boxed{F(z; x, u) = W(au)}. \quad (5)$$

In particular, objects with first list entry r have series w_ra^r . These identities hold as formal series with polynomial coefficients in $x, w_1, w_2, \dots$; only finitely many weights occur in each coefficient.

Proof. Temporarily leave the first list entry unweighted. This avoids ever dividing by a weight. Let $R(u)$ count these partially weighted trees, with u marking the first entry, and split it into $R_0(u)$ and $R_1(u)$ according as the root has no leaf child or has one. The empty tree belongs to R_0 . Let D count such trees after assigning their first entry of length r the weight w_{r+1} instead.

When L is nonempty, the first entry of the whole list is the first entry of L . The right list gains one at its beginning, so its completed weight is counted by D . When L is empty, the first entry remains unweighted and only increases by one. The two cases of the tree cut give

$$R_0(u) = 1 + zxDR_1(u), \quad R_1(u) = zD(R_0(u) - 1) + zuR(u).$$

The extra x in the first equation records the marked vertex lost in the second tree-cut case. Eliminating the two root states yields

$$R(u) = 1 + auR(u), \quad a = \frac{z(1 + zxD)}{1 - z^2xD^2}.$$

All denominators have constant term one. Therefore $R(u) = 1/(1 - au)$: the first entry r contributes simply a^r before it receives its weight. Restoring that weight proves (5), and also gives $D = W_+(a)$. Rearrange the displayed expression for a as

$$a - z = xz^2D(1 + aD).$$

Since $1 + aW_+(a) = W(a)$, this is (4). Finally the factor z^2 in its right-hand correction determines successive coefficients uniquely. This proves existence and uniqueness on the formal branch used by the combinatorial construction. $\square$

This is the natural multiplicative boundary of the argument: a run may receive any weight depending on its length, rather than just its parity. It also permits forbidden run lengths by setting the corresponding weights to zero. The first run has been retained throughout, so such specializations do not lose the boundary information needed by the cut.

4 Consequences with direct combinatorial meaning

Packing blocks of any length. For an integer $d \geq 2$, the maximum number of disjoint consecutive increasing blocks of length d in a permutation is $\sum \lfloor r/d \rfloor$ over its ascending runs. The intrinsic tree statistic $\sum_{r \in \lambda(T)} \lfloor r/d \rfloor$ therefore extends the original statistic to every d at once. Take $w_r = y^{\lfloor r/d \rfloor}$. Then

$$W(s) = \frac{1 + s + \cdots + s^{d-1}}{1 - ys^d}, \quad F(z; x, u) = \frac{1 + au + \cdots + (au)^{d-1}}{1 - y(au)^d},$$

where a is specified by (4). For $d = 2$ this gives the original joint statistics together with the first inverse ascending run.

The complete distribution of run lengths. If marked vertices are not weighted, put $x = 1$. Cancellation in the proof gives $a = z/(1 - zD)$, or equivalently

$$a = zW(a), \quad F(z; 1, 1) = a/z.$$

Suppose there are m_r runs of length r , with $\sum rm_r = n$ and $\ell = \sum m_r$. Lagrange inversion gives the exact number of permutations with this run-length multiset:

$$\frac{n!}{(n + 1 - \ell)! \prod_r m_r!}. \quad (6)$$

Indeed, it is the coefficient of $\prod w_r^{m_r}$ in $(n + 1)^{-1} [s^n] W(s)^{n+1}$. Formula (6) is a classical Catalan run-count consequence; the weighted theorem additionally keeps the descending statistic x and the first-run variable u .

An unweighted boundary check. Taking every $w_r = 1$ gives $a(1 - a) = z$. For $n \geq r \geq 1$, Lagrange inversion in $F(z; 1, u) = 1/(1 - au)$ then gives

$$\#\{T : |T| = n, \lambda(T) \text{ begins with } r\} = \frac{r}{2n - r} \binom{2n - r}{n}.$$

The only tree with an empty list is the one-vertex tree. These formulas also check the endpoints of the construction without making them exceptions to the main theorem.

Sources and scope

The original question and the source construction are linked below. The new formulation retains the full intrinsic run list and allows arbitrary multiplicative run weights. It does not claim that the classical Catalan coefficient formulas are new.

References

- [1] Z. Lin, J. Liu and S. H. F. Yan, *Parity statistics on restricted permutations and the Catalan–Schett polynomials*, Problem 2.18, arXiv:2409.01558v1.
- [2] EULER, Catalan–Schett plane-tree statistic.